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11/13 Based on lecture by: Marco Cuturi

Optimal Transport & Flow Matching

1 Motivating Examples

Optimal Transport can be viewed as the task of solving a puzzle about how to remap resources from one setting to another. Many natural phenomena can be interpreted through these puzzles, and OT provides a plausible and often elegant way to solve them.

1.1 Particle Tracking Toy Example

Imagine turning on a light and observing a configuration of particles. The light then turns off (perhaps the measurement device stops working) and during this period the particles move. When the light turns back on, we see a new configuration. Because particles are conserved, none have disappeared or appeared in the meantime.

It is useful to view this as two configurations in time and to ask how to connect them. There are many possible ways the particles could have moved, even under simple assumptions such as straight-line motion. The goal is to find a reasonable bijection that matches each particle in the initial configuration to one in the final configuration.

1.2 Single-Cell Microscopy Motivation

A similar situation occurs in single-cell microscopy. Imagine, we can observe the distribution of cells at two distinct times, but not their individual movements. Just like in the particle example, we want to connect these configurations and infer how the population transitioned from one to the other. This is a very real scenario often used on oncology research.

2 Optimal Transport v0.0: Linear Assignment Problem

As a first approximation, Optimal Transport can be viewed through the lens of the classical linear assignment problem. Suppose we observe N particles at an initial time and N particles at a later time. There are $N!$ possible bijections between these two configurations; even for $N = 6$, this already yields 6! possible pairings.

To make the matching problem meaningful, we introduce a cost function $c(x, y)$ that quantifies the expense of mapping a point x in the source configuration to a point y in the target configuration. The goal is then to identify the bijection that minimizes the total cost. Algorithms such as the Hungarian method are designed to solve this linear assignment problem exactly.

However, this formulation has several limitations. It does not scale well to large datasets, it is not particularly flexible, and it assumes discrete points with fixed masses in matching spaces. For practical applications, most users instead rely on the Sinkhorn algorithm, which provides a computationally efficient and differentiable approximation to optimal assignments.

Even so, two key challenges remain. *First*, the method still faces scaling issues in very large settings. *Second*, it does not naturally allow for out-of-sample extension: if a new point appears in the source distribution, the model does not directly specify where it should be mapped in the target distribution.

2.1 Applications in Single-Cell Omics

These issues become especially apparent in single-cell omics. A motivating example comes from tumor biopsies, where each cell is represented in an extremely high-dimensional gene-expression space (often $d \approx 35{,}000$). After administering a cancer drug, a second biopsy reveals a shifted distribution of cellular states. The central scientific question is: for each original cell, where did it “go” in the post-treatment population?

This setting has been studied extensively in recent work on applying Optimal Transport to cellular dynamics. For instance, Peyrl et al. (2025) introduce *moscot*, an OT-based framework for mapping cellular trajectories through time [PPL$^+$25]. Similarly, Cazelles et al. (2023) present a comprehensive review on Optimal Transport for single-cell and spatial omics [CDH$^+$23]. Both works highlight the need for transport maps that can be evaluated on arbitrary new points rather than only on the discrete samples used during training.

In this context, classical linear assignment or Sinkhorn-based couplings are not sufficient: beyond scalability concerns, they lack an out-of-sample extension mechanism. What is needed is a continuous map capable of pushing forward the entire source distribution to the target distribution, enabling predictions about how unseen or hypothetical cells would respond to perturbations such as drug treatment.

2.2 Applications to Generative Modeling

Optimal Transport also provides a useful perspective on modern generative modeling. One can view the generative task as learning a map that pushes forward a simple source distribution—typically random noise—to a complex data distribution supported on a particular class of images. Again, in this scenario, it is important to learn a map as opposed to an assignment between fixed points.

3 OT Toolbox: Monge Maps and Potentials

Given the limitations of linear assignment, we motivate the traditional OT formulations learned previously in the course. We redefine them here for reference.

3.1 Monge Maps

A Monge map is a transport map $T : \mathbb{R}^d \to \mathbb{R}^d$ that pushes a source measure μ onto a target measure ν. In the picture, μ (red) is transported to ν (blue) via T, and sets B in the target space correspond to preimages $T^{-1}(B)$ in the source space.

$$T_{\#}\mu = \nu \quad \Longleftrightarrow \quad \forall B \subset \mathbb{R}^p,\ \nu(B) = \mu\big(T^{-1}(B)\big).$$

The optimal Monge map T^* is defined as the minimizer of the total transport cost:

$$T^* = \underset{\substack{T:\mathbb{R}^d \to \mathbb{R}^d \\ T_{\#}\mu = \nu}}{} \int c\big(x, T(x)\big)\, d\mu(x).$$

For example, we can define the quadratic cost

$$c(x, y) = \frac{1}{2}\,\|x - y\|^2.$$

3.2 Kantorovich Formulation and Dual

The Monge formulation requires a deterministic transport map and therefore does not allow any splitting of mass: each point x must be sent entirely to a single point y. In many applications this is too restrictive, so the Kantorovich relaxation introduces a more flexible notion of transport by allowing joint couplings.

The Kantorovich problem seeks a coupling P between μ and ν:

$$P^* = \arg \inf_{P \in \Pi(\mu,\nu)} \iint c(x, y)\, P(dx, dy),$$

where $\Pi(\mu, \nu)$ denotes the set of all joint distributions with marginals μ and ν.

Although this relaxation makes the problem well-posed and far more general than Monge, the Kantorovich formulation is not computationally tractable for large-scale problems. This motivates turning to the dual problem.

The dual problem is:

$$f^*,\, g^* = \arg \sup_{f,g:\mathbb{R}^d \to \mathbb{R}} \left(\int_{\mathbb{R}^d} f \, d\mu \; + \; \int_{\mathbb{R}^d} g \, d\nu \right) \quad \text{subject to} \quad f(x) + g(y) \le c(x,y).$$

This dual perspective is used extensively in modern Optimal Transport, both for theoretical analysis and for practical algorithms.

3.3 Monge–Ampère Equation and Brenier's Theorem

When the cost function is the quadratic cost

$$c(x,y) = \|x - y\|^2,$$

the structure of the optimal transport map becomes significantly more rigid. In this setting, one can prove Brenier's Theorem, which states:

$$\text{Brenier's Theorem: if } c = \|\,\cdot - \cdot\,\|^2, \text{ then } T^*(x) = \nabla f^*(x), \quad f^* \text{ convex.}$$

If the measures admit densities, written as

$$\mu = p\,L, \qquad \nu = q\,L,$$

then the pushforward condition $T_*\mu = \nu$ becomes

$$T_*\mu = \nu \iff p(x) = q\big(T(x)\big)\,\big|\det J_T(x)\big|.$$

Using the form $T^* = \nabla f$, this leads to the classical Monge–Ampère equation, in which one seeks a convex potential f satisfying

$$|\nabla^2 f(x)| \;=\; \frac{p(x)}{q\big(\nabla f(x)\big)}.$$

The Monge–Ampère formulation is crucial both for theoretical understanding and for developing algorithms that exploit convexity and PDE structure, and it serves as a bridge between the geometric, analytic, and computational aspects of optimal transport.

4 Many Facets of Optimal Transport

At this stage of the lecture, Cuturi emphasized that Optimal Transport does not come in a single form: it has many different facets, each offering a distinct way to view and formulate transport problems. Depending on the context and the type of data one works with, different formulations become appropriate and lead to different computational tools. In some applications, a simple linear assignment formulation may already be sufficient; in others, more flexible or more geometric formulations are required.

Beyond the Monge, Kantorovich, and dual formulations discussed earlier, there are numerous additional perspectives that extend Optimal Transport in different directions. Examples include Schrödinger bridges, multi-marginal OT, and quadratic OT. These perspectives broaden the applicability of OT by capturing uncertainty, temporal dynamics, or complex multi-way interactions.

4.1 Best Formulation of OT?

During the lecture, Cuturi was asked which formulations are overused and which are likely to become more relevant in the future. He remarked that when Brenier's formulation first appeared, he initially thought it would be useless—only for it to later become central to the field. From this experience, he drew the conclusion that the long-term relevance of an OT formulation is tied not to its mathematical elegance or complexity, but to whether it ultimately becomes computationally viable. In practice, computational efficiency determines which formulations gain widespread use and which remain primarily theoretical.

5 Static vs. Dynamic Generative Formulations

With this larger toolbox establish, we revisit the topic of generation models.

5.1 Static Monge Map (One-Step Generation)

A static formulation treats generation as a single transformation: a map that pushes a simple source distribution (e.g., noise) directly to the data distribution in one step. This is analogous to one-step generative models such as GANs, where a single deterministic function $T(x)$ produces a sample in one forward pass.

5.2 Dynamic Benamou–Brenier (Multi-Step Generation)

In contrast, dynamic formulations view generation as a multi-step process. The Benamou–Brenier perspective represents transport as a continuous-time flow connecting the source and target distributions. This viewpoint is similar to multi-step generative approaches such as Neural ODEs, continuous normalizing flows, and, in spirit, diffusion models. These methods construct the final transformation by composing many small increments, which can offer more stable learning and help capture more complex structure. We will discuss the Benamou–Brenier formulation in more detail in the next section.

6 Benamou–Brenier Velocity Formulation

This is a more *dynamic* viewpoint on optimal transport. Instead of directly searching for a static map that pushes μ_0 to μ_1, the idea is to let the distribution evolve smoothly over time through a velocity field. We discuss a family of distributions $\{\mu_t\}_{t\in[0,1]}$ and a velocity field $v_t(x)$ such that particles move according to

$$x_{t+\Delta t} = x_t + \Delta t \cdot v_t(x_t),$$

with boundary conditions μ_0 (initial distribution) and μ_1 (final one).

The change in density follows a mass conservation idea: the time derivative of μ_t at a location depends on mass flowing in and out due to the velocity field. OT then selects the path that minimizes the kinetic energy, so among all possible ways to morph μ_0 into μ_1, we choose the most "efficient" one.

One issue is that the original problem is not convex in (μ_t, v_t). A common trick is to reparameterize by defining

$$\rho_t = v_t\, \mu_t,$$

which turns the optimization into a convex problem over (μ_t, ρ_t). This is helpful in high dimensions and is the basis behind practical solvers like OTT-JAX.

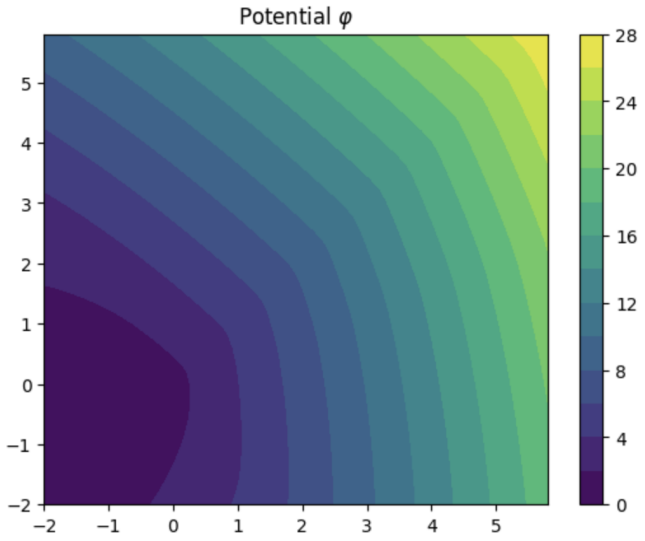


Figure 1: Convex Brenier potential φ whose gradient generates the optimal velocity field. (Adapted from OTT-JAX tutorial.)

7 Practical Tools: OTT-JAX and Brenier Potentials

We briefly looked at how these ideas show up in practice, especially in OTT-JAX, which has support for both static OT and Benamou–Brenier dynamic flows.

Relevant tutorials:

- Benamou–Brenier flow theory + animations: `https://ott-jax.readthedocs.io/tutorials/theory/000_bb_flow.html`
- Brenier potential → velocity field: `https://ott-jax.readthedocs.io/_autosummary/ott.math.velocity_from_brenier_potential.html#ott.math.velocity_from_brenier_potential`

Some key ideas shown through their animations:

- **McCann interpolation**: smooth barycentric interpolation between μ_0 and μ_1.
- **Velocity from convex potentials**: gradients of a convex potential give the transport directions.
- **BB integration demo**: mass moves along geodesics in Wasserstein space.

8 Flow Matching and OT-FM

Flow matching (FM) gives another way to learn velocity fields for distribution transport. Instead of solving OT directly, we define an interpolation μ_t and train a model whose velocity matches the infinitesimal displacement.

8.1 Independent Flow Matching (IFM)

A simple interpolation is:

$$X_t = (1-t)X_0 + tX_1,$$

where $X_0 \sim \mu_0$ and $X_1 \sim \mu_1$ are sampled independently. Because the pairs are not matched in any geometric sense, there are many possible paths between points, and small batches make this especially unreliable.

The IFM objective shown in class was:

$$\min_{\theta} \ \mathbb{E}_{T\sim U[0,1],\, X_0\sim\mu_0,\, X_1\sim\mu_1}\left\|X_1 - X_0 - v_\theta(X_T, T)\right\|^2, \qquad \text{where } X_T = (1-T)X_0 + TX_1.$$

This formulation treats sample pairs independently and can lead to noisy flow fields that do not respect the geometry of the target distribution.

A relevant reference mentioned on the slide was:

> *Multisample Flow Matching: Straightening Flows with Minibatch Couplings* (Pooladian et al., 2023).

8.2 OT-Enhanced Flow Matching (OT-FM)

OT-FM fixes that by pairing samples using optimal transport first:

- For each batch, solve an OT coupling between source and target.
- Interpolation paths become straighter and better aligned with actual transport.

Limitations:

- mini-batch OT can create misleading correspondences when the batch is too small, causing poor gradient updates;
- as $\varepsilon \to 0$, the mapping becomes sharp (Hungarian-like), reducing distributional smoothing.

8.3 Large-Scale OT-FM and Regularization

The lecture emphasized that performance improves significantly when we use *large* OT batch sizes and *small* regularization ε:

$$\varepsilon \to 0 \Rightarrow \text{sharp / Hungarian-like}, \qquad \varepsilon \to \infty \Rightarrow \text{independent sampling}.$$

Compute cost note: Sinkhorn is roughly $O(dn^2)$ but OTT-JAX can shard computation across GPUs.

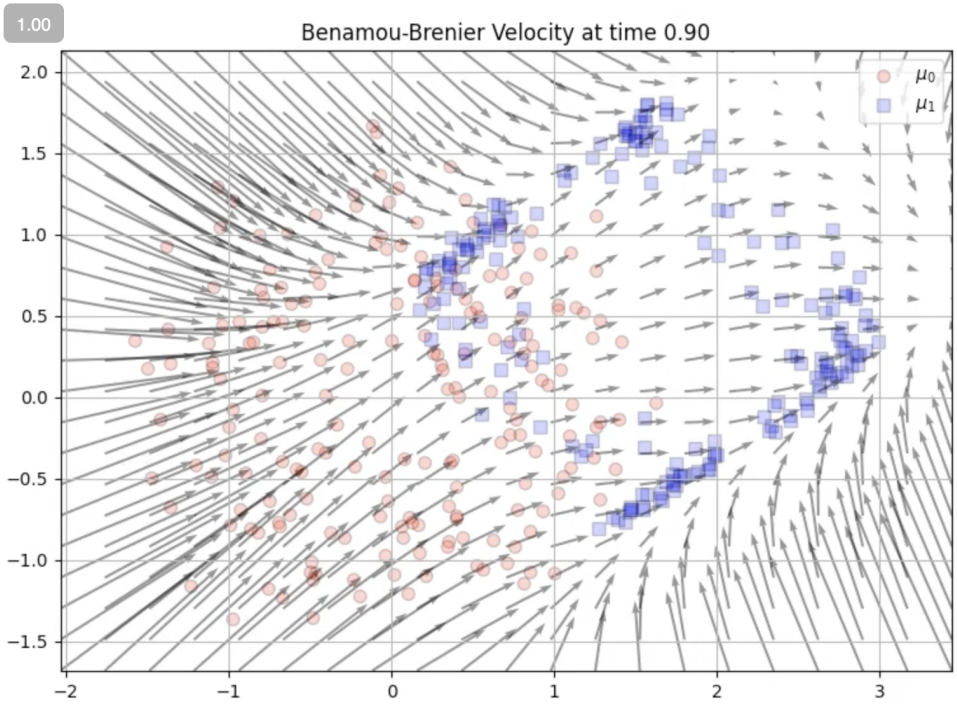


Figure 2: Benamou–Brenier velocity field showing smooth transport from μ_0 (red) to μ_1 (blue). (Adapted from OTT-JAX tutorial.)

9 Results: Large Batch, Small ε and Semidiscrete OT

This part of class summarized experiments on synthetic data + ImageNet-64 evaluating I-FM vs OT-FM.

These comparisons follow the setup from Pooladian et al. (2023) and Tong et al. (2023), where minibatch OT couplings are added to flow matching.

Key findings:

- OT-FM significantly outperforms I-FM.
- Larger OT batch sizes B (e.g., 10^3–10^6 points) improve results.
- The gap is most noticeable when inference needs to be cheap/low-cost.

9.1 Motivation for Semidiscrete OT

Even with GPU Sinkhorn, cost can be high:

$$\text{Sinkhorn complexity } \approx O\left(\frac{dn^2}{\varepsilon^2}\right).$$

So the next idea: use **semidiscrete OT**, where one distribution is continuous (noise) and one is discrete (dataset). In the generative setting:

$$\text{noise is continuous, data is finite of size } N.$$

Message from lecture:

$$\text{“infinite } n\text{” meaning } n = N, \quad \text{and can also set } \varepsilon = 0.$$

9.2 Semidiscrete Flow Matching (SD-FM)

The slide compared three setups:

Method	Quick summary
I-FM	Independent pair interpolation; tends to produce noisy flow fields with less geometric structure.
OT-FM	Uses optimal transport pairings before interpolation; produces straighter, more meaningful transport paths.
SD-FM	Precomputes a semidiscrete dual potential g and assigns noise samples via $\arg\max_j(x^\top y_j + g_j)$, improving stability and smoothness.

Table 1: Comparing Independent Flow Matching (I-FM), OT-Enhanced Flow Matching (OT-FM), and Semidiscrete Flow Matching (SD-FM).

The semidiscrete training step uses:

$$j^* = \arg\max_j (x^\top y_j + g_j),$$

with loss (for $\varepsilon > 0$):

$$\ell_\varepsilon(g, x_0) := -\sum_{i=1}^N g_i \nu_i + \varepsilon \log\left[\sum_{i=1}^N \exp\left(\frac{g_i - c(x_0, x_i^1)}{\varepsilon}\right)\nu_i\right],$$

and for $\varepsilon = 0$:

$$\max_i g_i - c(x_0, x_i^1).$$

9.3 Algorithms (Semidiscrete OT)

Two short algorithms were mentioned for the semidiscrete OT setup. These are not full training loops, but the basic routines used during SD-FM.

Algorithm 1: SolveSDOT (computes dual potential g). The goal is to estimate a vector $g \in \mathbb{R}^N$ over the dataset points $\{x_i^1\}_{i=1}^N$.

- Initialize $g \leftarrow 0$.
- Repeatedly sample noise points $x_0 \sim \mu_0$.
- Update g by descending the semidiscrete OT loss:

$$\ell_\varepsilon(g, x_0) := -\sum_{i=1}^N g_i \nu_i + \varepsilon \log\left(\sum_{i=1}^N \exp\left(\frac{g_i - c(x_0, x_i^1)}{\varepsilon}\right)\nu_i\right),$$

  and in the $\varepsilon = 0$ case,

$$\ell_0(g, x_0) = \max_i \left(g_i - c(x_0, x_i^1)\right).$$

- After some iterations, return $\tilde{g} \approx g$.

Algorithm 2: Couple (assigns noise data). Given a noise sample x_0 and the potential g:

- Compute scores: $z_j = g_j - c(x_0, x_j^1)$.
- If $\varepsilon > 0$, sample using

$$p_j = \text{softmax}(z_j/\varepsilon);$$

  otherwise return

$$j^* = \arg\max_j z_j.$$

- Output the pair $(x_0, x_{j^*}^1)$.

Why this helps. Unlike OT-FM, which repeatedly solves mini-batch OT, SD-FM reuses the potential g during training, which can make it more scalable when the dataset is large.

9.4 Qualitative Visual Comparisons

We also saw plots of velocity fields:

- IFM after 10k iterations: flow is noisy and less aligned.
- OT-FM ($n = 128, \varepsilon = 0.01$) after 10k iterations: noticeably cleaner flow.
- OTT-JAX demo showed Brenier velocity on ring-shaped distributions.

9.5 Main Takeaways

- OT-FM improves significantly when batch size n is large and ε is small.
- Semidiscrete OT gives an additional edge in the generative setting (continuous noise + finite data).
- Practical scaling relies on distributed Sinkhorn + GPU support.
- Overall trend: structured OT pairings lead to smoother, more meaningful flows.

Acknowledgments

Some illustrations are adapted from the OTT-JAX documentation and Benamou–Brenier flow tutorial.

References

[CDH+23] Charlie Cazelles, Michael Dorrity, Michael Heiser, et al. Optimal transport for single-cell and spatial omics. *Nature Reviews Methods Primers*, 2023.

[PPL+25] Dominik Peyrl, Giovanni Palla, Marco Lange, Tapio Lönnberg, et al. Mapping cells through time and space with moscot. *Nature*, 2025.